XIAMEN UNIVERSITY MALAYSIA

TEST (20%)

Course Code:	COM103		
Course Name:	Electronic Engineering		
Question Paper Setter:	Assoc. Prof. Ir. Ts. Dr. Wong Shen Yuong		
Academic Session:	2022/04	Question Paper:	A ☑
Total No. of Pages:	2	Time Allocated:	1 hour
Additional Materials:	-		
Apparatus Allowed:	Calculator		

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 2 Questions.
   Answer **ALL** questions.
2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirement of each section and write down all the corresponding answers on the lined paper.
4. Only PDF hand written answer sheet will be accepted.
5. Submit your answer sheet on Moodle link "Test-Submission", which there will be a cut-off time.

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID: _____ Full Name: _____)

Question 1 [13 marks]

(CLO1, PLO1, C2)

(a) Comparing semiconductor and conductor, which one has more free electrons? Why ? [2m]

(b) Explain what happen to semiconductor if there is an increase in temperature? [2m]

(c) Figure 1 shows a diode circuit connected to a direct current (DC) voltage source V_1 of 5 V and a resistor R_1 of 2 kΩ in series. Solve the circuit using the 'Complete Diode' model. Assume it is a silicon diode D_1, r'_d = 10 Ω determine the voltage across diode (V_{D1}) , forward current for the diode (I_F), and the voltage drop across resistor R_1 (V_{R1}). [6m]

(d) What if the diode is reversed bias in Figure 1? Determine the voltage across diode (V_{D1}) , forward current for the diode (I_F), and the voltage drop across resistor R_1 (V_{R1}). [3m]

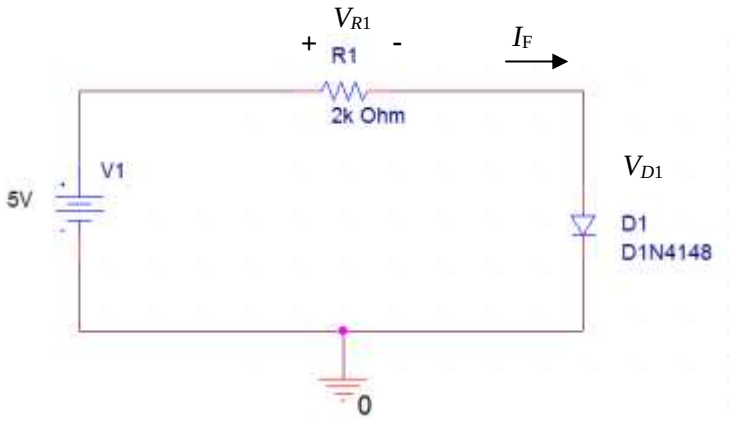


Figure 1

Question 2 [7 marks]

(CLO1, PLO1, C3)

Referring to Figure 2,

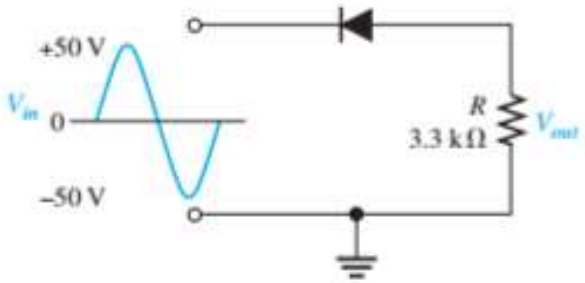


Figure 2

(a) Analyse and draw the output voltage waveform. [2m]

(b) What is the PIV across each diode? [1m]

(c) Calculate the average value of a half-wave rectified voltage with a peak value of 100V. [2m]

(d) What is the peak forward current through the diode? [2m]

- End of Question-